

Hints for the Final Exam

December 13, 2010

We will use two benchmark models: the binomial models and the Black-Scholes model. No arbitrage will be a standard assumption unless otherwise stated. Here are some points you need to pay attention.

- Know how to price an European call and a lookback option in a two-period binomial model. Both the backward recursion and the risk-neutral expectation are required. In Black-Scholes model, you can derive the Black-Scholes PDE for European call price, and you know Black-Scholes formula.
- Know how to price a forward contract with risk-neutral expectation, and be able to use Put-Call parity. Notice: put-call parity is only true for European options.
- Know how to price an American put/straddle in a two-period binomial model using backward recursion.
- For the above American derivatives, be able to find the optimal exercise time for each contract.
- The physical probabilities (p, q) in binomial models, and the physical mean rate of return (α) in Black-Scholes model, are not relevant in derivative pricing, because risk-neutral measure is the correct measure for derivative pricing.
- In BS model, stock A has an annualized mean rate of return $\alpha_1 = 30\%$ and volatility $\sigma_1 = 10\%$, stock B has an annualized mean rate of return $\alpha_2 = 15\%$ and volatility $\sigma_2 = 20\%$. Assume both stocks are worth 100 now, compare the prices of co-terminal at-the-money calls on these two stocks.
- Is it ever optimal to early exercise an American call option?
- A put and a call on a stock struck at the forward price have the same value by put-call parity. Yet the value of a put is bounded and the value of a call is unbounded. Explain how they can have the same value?
- Which is riskier: a call or the underlying?
- Do you gain anything over the risk-free rate by continuously delta-hedge a call option in a ideal B-S model?
- You can solve for Vasicek model and work out the mean and variance.
- Assume we have a risk-neutral measure (so no arbitrage is allowed), and the underlying price process is continuous (not necessarily Black-Scholes). The interest rate is always zero, and the present stock price is 100, what is the price of a up-and-out put struck with strike at 120 and knockout barrier at 120?
- Interpretation of Black-Scholes formula: the time-0 price of a call struck at K is given by $S_0 N(d_+) - Ke^{-rT} N(d_-)$. Show that $N(d_-)$ is the risk-neutral probability that the underlying will finish in the money. Construct another probability $\tilde{P}^S$ and show that
$$N(d_+) = \tilde{P}^S(\text{"the underlying will finish in the money"}).$$
- Sketch the graph of delta, gamma and theta of an European call.
- Assume that there exists a risk-neutral measure. Also assume the underlying process is continuous (but not necessarily B-S model). Consider a digital option that pays \$1 if the price increases to 200. What is the present price of the option if the current stock price is 100.
- What is the distribution of $\int_0^1 W_t dt$, where $\{W_t\}$ is a standard Brownian motion.